

ARTICLE

How do students use ChatGPT as a writing support?

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INTRODUCTION

In many out-of-school settings, writing is a collaborative activity. Writers plan with audiences in mind, generate ideas with colleagues or co-authors, draw from models as they organize their ideas and translate them into sentences, and review their work with the help of human and digital editors. These distributed and “dialogic processes of invention” (Prior, 2006, p. 58) can make for stronger writing (Graham et al., 2023).

Historically, however, schools have framed writing as a more “sequestered” activity (Klein & Leacock, 2012), in which students brainstorm, draft, and revise mostly on their own. In this sequestered view of writing, a student would have to engage in thousands of hours of individual practice to gain writing expertise (Ericsson et al., 1993). And because writing instruction is so time-consuming and challenging, students do not get anywhere near that kind of practice in school. For instance, when Applebee and Langer examined 8000 written assignments from 130 students in U.S. middle and high schools, they found that about 80% of students’ writing comprised short answer and fill-in-the-blank work—“types of activities that are best described as writing without composing” (Applebee & Langer, 2011, p. 15). Other national data show that students also have limited practice in the key writing process of revision (Kiuahara et al., 2009).

So when ChatGPT became widely available in 2022, many teachers responded first with concern. On top of existing worries about misinformation, bias, and privacy (Rawas, 2024), teachers and administrators expressed worry that students would use ChatGPT as a cheatbot—a way to sidestep the difficult practice of learning to write (e.g., Cotton et al., 2023). Their fears were likely compounded by a wave of news stories about student cheating with ChatGPT (e.g., McDade, 2023). At the same time, teachers learned that conventional plagiarism detectors could not reliably identify text-based AI in student writing (Chaka, 2023).

Since 2022, schools, teachers, and researchers have begun to reconsider their assumptions about cheating and generative AI tools. Schools have reversed bans on ChatGPT, and teachers have experimented with integrating ChatGPT into writing instruction (Banks, 2023; Youth Voices, 2024). Schools have also worked to develop plagiarism policies to accommodate the many uses of generative AI (Barnett, 2023), and researchers have found that students and teachers differ in their definitions of cheating with chatbots like ChatGPT (Mah et al., 2024). For example, is it cheating to get ideas from ChatGPT, or is it only cheating if you use ChatGPT’s language to articulate those ideas? Meanwhile, to the degree that people can agree on a definition, a national survey showed that students did not self-report cheating with ChatGPT anymore than they did before ChatGPT (Lee et al., 2024).

While many maintain justifiable concerns about AI’s potential role in diminishing students’ attention to writing, many have also speculated about its potential benefits. ChatGPT could, for example, act as a writing coach or muse (Kleiman, 2022) or “support bottom-up writing skills, thus leaving more time, space and energy to be devoted to the higher pursuits of composition” (Daniel et al., 2023, p. 37). Mollick and Mollick (2023) offer an inventory of ways that students can position AI as a partner in their work, e.g. as a mentor, coach, teammate, tutor, or as a fellow student (in that last role, one would try to teach the AI, and in doing so, have to clarify concepts and processes for oneself). It is even possible that AI might allow not just for augmentation of writing processes, but for modifying or even redefining writing (Puentedura, 2013), just as Google Docs arguably redefined writing by allowing co-writers to simultaneously collaborate across locations.

All these speculations help guide the field of literacy education away from the traditional sequestered model of writing and towards a model of writing as an unsequestered, social activity that is distributed across many resources, both human and digital (Klein &

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Leacock, 2012). However, beyond these speculations, literacy researchers and educators have little empirical evidence about how students productively engage AI in this unsequestered model.

Thus, we designed a classroom-based study not to explore cheating with tools like ChatGPT, but instead to explore how students might use such tools as part of a distributed writing model to generate ideas, organize, draft, or evaluate their writing.

We investigated how a group of 12 high school students used ChatGPT as they wrote argumentative essays. Specifically, we asked:

1. In terms of planning, translating, and evaluating, what did students ask ChatGPT to do?
2. How did students take up ChatGPT's responses to their prompts?

AI AS PART OF A SOCIAL MODEL OF WRITING

Writing is social activity

In their foundational cognitive models of writing, Flower and Hayes (1981) broke down the writing process into several subprocesses: *planning* (generating ideas, setting goals, and organizing), *translating* (text generation and moving from ideas in the head to generating actual text on a page or screen), *reviewing* (reading, evaluating, revising, and editing), and *monitoring* (coordinating all of the above processes for optimal performance). Although cognitively oriented, this and related models acknowledge that writing does not take place solely within the head of an individual writer. For instance, Flower and Hayes point out that a writer's "task environment"—their understanding of their audience—influences all aspects of a writer's process. However, these models tend to highlight the individual writer's work of integrating and monitoring different writing processes. In this way, such models support a view of writing as a sequestered activity (Klein & Leacock, 2012).

More recently, however, teachers and researchers have presented more expansive models of the writing process, arguing that writers operate not alone, but within communities (Graham, 2018). Klein and Leacock (2012) argue that writing is essentially a case of distributed cognition. They review ways that students distribute the planning, translating, and reviewing across many familiar platforms and resources. For instance, a teacher often plays a significant role in helping students *plan* their writing, designating writing topics and offering rubrics. Students may *translate* ideas to sentences using graphic organizers or sentence frames (Levine, 2019). As students *review* their work, they may check their spelling and grammar with digital applications.

When students do draw on outside support, their writing improves. A meta-analysis of 406 intervention studies (Graham et al., 2023) tracked the reported effects of instruction that unsequestered the writing process, finding that peer assistance, teacher feedback, or computer-assisted instruction resulted in richer idea generation, stronger organization, and increased writing knowledge and motivation.

Now, with the emergence of ChatGPT, Bard, and other text-based generative AI tools, our field can study students' unsequestered writing in new ways.

Adding AI to social activity

Researchers are beginning to evaluate the role of chatbots as part of the distributed model of writing. Many studies suggest that role is promising (e.g., Su et al., 2023). In an experimental study with more than 300 university students, Lin and Chang (2020) found that students who interacted with an AI-driven chatbot wrote stronger thesis statements and earned higher scores on an essay outline assignment compared to students who did not. But other studies have yielded less encouraging results. For example, Bašić et al. (2023) compared students' writing with and without AI assistance, finding that graduate students who wrote with ChatGPT did not write more or create more high-quality content than those who wrote without ChatGPT.

Other research shows promise in automated writing evaluation (AWE) software, which offers feedback for revision of grammar, language, and diction. In a meta-analysis of fifty-eight studies, Ngo et al. (2022) concluded that AWE systems could be beneficial writing tools for a wide range of student populations, including students with intermediate English proficiency. For instance, engaging with AWE feedback can help students monitor and evaluate their own writing (Zhang, 2020). However, other studies have found that many students do not choose to use AWE applications, or that when they do, they use them mostly for surface level edits (Warschauer & Grimes, 2008).

Researchers have now begun to examine the potential of ChatGPT as a feedback provider. For instance, researchers Su et al. (2023) wrote an argument in favor of shorter school days and submitted it to ChatGPT for evaluation and editing. They found that ChatGPT accurately identified strengths and weaknesses in their argument, was responsive to their requests for counterarguments, and effectively edited their work. Dai et al. (2023) compared the quality of ChatGPT-generated and human feedback to a set of graduate students' business proposals, finding ChatGPT's feedback to be more readable and encouraging than that of human readers. Gero et al. (2023) found that ChatGPT was able to tailor feedback to

creative writers; for instance, ChatGPT suggested possible endings to a piece of fiction.

These studies illustrate the potential of AI as a support for writing. However, many of these studies focus on university students and adults in one-shot interactions with text-based AI. Researchers and teachers need more empirical work on high school students' productive interactions with these tools across planning, translating, and evaluating.

STUDY DESIGN

Participants and setting

To conduct this research, we partnered with a teacher at a California charter high school, which we will call Sunrise High. Sunrise's mission is to help first-generation, low-income students go to college. The school serves Black, Latinx, and AAPI students who are all eligible for free or reduced lunch.

The teacher and the Sunrise administration were interested—and anxious—to better understand how their students might use ChatGPT, and the degree to which students might see ChatGPT as a way to avoid writing processes. Our partner teacher helped us recruit 12 students who were able to meet in after-school sessions to do some writing with ChatGPT. The group included one 9th grader, five 10th graders, five 11th graders, and one 12th grader. All students knew one another and the teacher. We offered the students and teacher a stipend for their time.

Our university's institutional review board approved this study. Students, guardians, and teacher gave informed consent to participate.

Methodology

To the extent possible in these after-school sessions, our goal was to capture “naturally occurring” student activity (Gabrielsen et al., 2019) as students worked with ChatGPT to plan, translate, and review their writing. In addition to designing an activity to introduce ChatGPT to all students, we designed two writing prompts that mimicked typical school argument assignments, such as a prompt about whether Sunrise High should institute a school uniform policy. Once we gave students their assignment for the writing sessions, we did not sit with them while they wrote or prompt them in any way.

Data collection

Over two weeks, the research team and partner teacher facilitated two sixty-minute after-school sessions during

which students used Open AI's ChatGPT 3.5 to complete these argument writing tasks. After each session, they shared their final arguments with us.

In the first session, we asked students to compose with an assigned partner, so that we could use their dialogue to gain insight into their writing choices and their assessments of the value of ChatGPT. Students responded to the following written argument prompt:

Imagine that the Sunrise administration is considering adopting a new mascot to represent Sunrise. They are considering these three mascots: Redwoods; Seals; and Warriors. The school administration is asking students to submit their arguments for and against each of these mascots. Alternatively, students can suggest their own idea for a mascot.”

In your argument, the administration wants you to include:

- An opening that sets the stage for your argument
- Your claim about the mascot that best represents Sunrise
- Explanation and specific examples that support your claim
- Your claims about why each of the other choices do not best represent Sunrise
- Explanation and specific examples to support your claim
- An interesting conclusion that explains why this matters to you and why it should matter to the administration
- Careful editing

In the second session, students worked without writing partners. We asked students to make arguments for or against instituting school uniforms or year-round school calendars.

In both sessions, students used Chromebooks to access ChatGPT and Google Docs, along with any other digital tools they wished. Our specific instructions were: “Please use ChatGPT to help you, along with anything else you would normally use to write. You can use ChatGPT any way you want, except to write your entire argument for you.” In addition to ChatGPT, students used other tools, including the spell-check feature of Google Docs, an online thesaurus, and websites that offered information about seals, redwoods, or other potential mascots.

Our data included:

- *Screencasts* (audio of students' talk and videos of their computer screens) of all sessions ($n=22$).

Screencasts allowed us to observe students' writing while hearing them speak to one another (or occasionally to themselves) about that writing. Screencasts also allowed us to note when students toggled between ChatGPT and their Google Doc, when they consulted other digital resources, and how much time they spent looking at ChatGPT output (see Figure 1 for an example). We used human and AI services to transcribe all student talk.

- *Transcripts* of students' written prompts to ChatGPT along with ChatGPT output ($n=22$).
- *Students' written texts* ($n=26$).

Data analysis

We employed different analytic approaches to explore our two questions.

What did students ask ChatGPT to do?

To better understand the processes students engaged in when prompting ChatGPT, we borrowed from Flower and Hayes' model, coding each prompt as belonging to one or more of the following categories: planning, translating, and reviewing. Three members of the research team collaboratively coded (Terry et al., 2017; Smagorinsky, 2008) all student prompts ($n=64$). Table 1 provides examples of each code.

How did students take up ChatGPT's output?

To explore how students used ChatGPT's output, we conducted both quantitative and qualitative analyses.

First, given many educators' concerns about ChatGPT's possible role in side-stepping writing, we used a computational statistic to measure the frequency with which students took up ChatGPT's language as they planned and translated their arguments. For instance, while planning their argument, one writer asked ChatGPT to offer data in favor of a year-round school calendar. The bot used the term "summer slump" in its response. The student then used that phrase in their final argument.

We used a computational text analysis statistic called Jaccard similarity (Wu et al., 2021), which measures the similarity between two groups of words (plagiarism detectors such as [turnitin.com](https://www.turnitin.com) use similar computational methods). Jaccard similarity measures the number of individual words that both texts share divided by the total number of words in both texts. For example, if a student's draft and ChatGPT's output were both a hundred words long, and the two texts had 50 words in common, Jaccard similarity for those two texts would be 25%.

We first calculated the percentage of shared individual words (for example, whether the word "mascot" appeared in a student's ChatGPT output and the student's subsequent draft). However, this measure offered only limited information. If "mascot" was the topic of argument, that word would likely appear across students' inputs and ChatGPT's outputs, and thus the measure would not tell us whether students took up ChatGPT's language. However, Jaccard similarity also allows for comparisons of two-word strings (e.g., "mascot represents" appears in both ChatGPT and essay), and three-word strings (e.g., "mascot represents strength"), so we calculated those as well.

To account for possible variations of word form (such as plural or singular nouns), we converted each word to what is called a lemma using Natural Language Toolkit

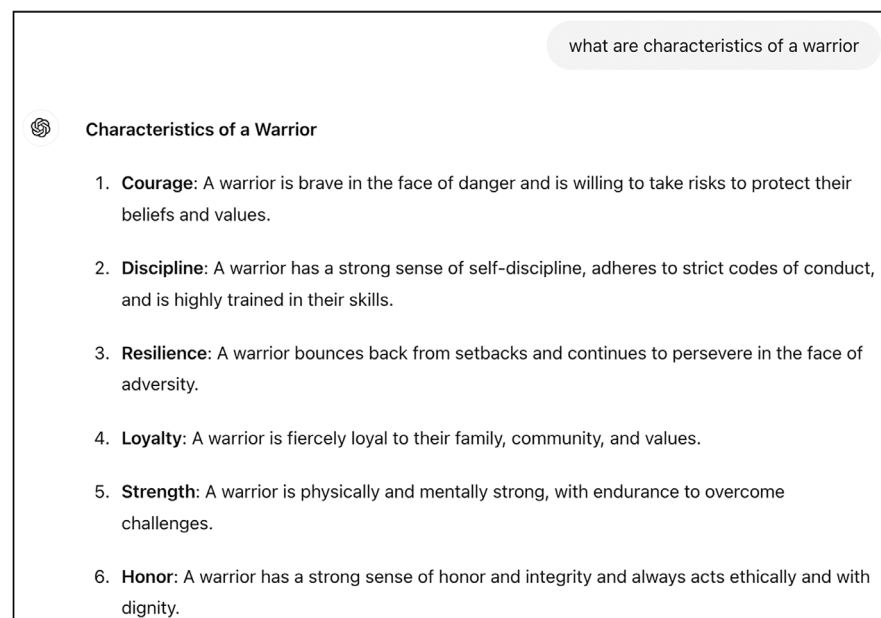


FIGURE 1 Student prompt and ChatGPT output captured during a screencast.

TABLE 1 Codes and examples of Students' prompts to ChatGPT.

Code	Code definition	Examples from students' prompts to ChatGPT
Planning	Any request connected to goal-setting, generating ideas, organizing	"Why would trees be a weak school mascot?" "What do warriors symbolize?"
Translating	Moving from idea generation to outlining or actually writing words and sentences	"Can you give me an example of a hook for argumentative essays regarding a change in school mascots?" "What's a word with a more positive connotation which means the same as <i>suitable</i> ?"
Reviewing	Requests related to reading, evaluating, and modifying students' writing	"Can you proofread this paragraph?" "Can you give me some stronger power words?"

(version 3.8). For example, "running," "ran" and "runs" would all be converted to the lemma "run."

This comparison was useful when comparing ChatGPT's output with students' evolving drafts. However, it was not useful when comparing students' completed drafts with ChatGPT's edits of those drafts. For example, in five instances, students copied and pasted their whole draft into ChatGPT and asked the bot to edit or proofread. In these cases, students' writing and ChatGPT's output were very similar, so we manually compared the students' drafts, ChatGPT's edits, and the final arguments that students ultimately submitted to us. This comparison allowed us to see if students had taken up ChatGPT's editing suggestions here and there, or if they took up all of the bot's edits in their final draft.

RESULTS

In this study, we explored how ChatGPT functioned as writing support in an unsequestered model of student writing. We analyzed students' input (what students asked of ChatGPT as they engaged in the writing process) and what students took up from ChatGPT's output.

What did students ask ChatGPT to do?

In total, the 12 students prompted ChatGPT 64 times. When writing with a partner, students prompted ChatGPT a total of 37 times (average of 6.1 times per pair of students). When writing without a partner, students prompted ChatGPT a total of 27 times (average of 2.7 times per student). The majority of prompts (52) involved requests that fell into the *planning* category. Twenty prompts included *reviewing* requests, and ten prompts included requests for *translating*. The total number of codes is greater than 64 because many prompts included requests in more than one category (See Figure 2 for percentages of each code).

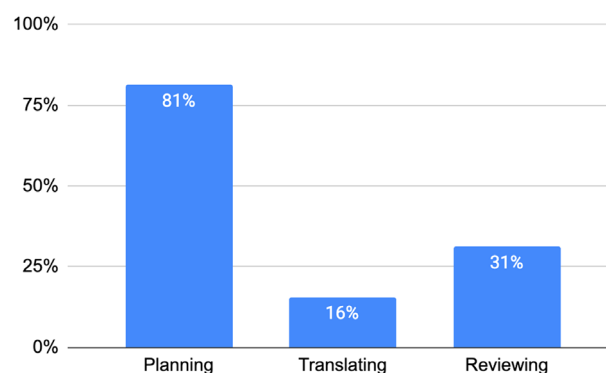


FIGURE 2 Percentages of prompts coded as planning, translating, and/or reviewing. Note: Prompts could receive more than one code, so percentages sum to more than 100%.

Planning

Most of the students' prompts fell into the planning category—*goal-setting*, *generating ideas*, and *organizing*. Many students asked for ideas, as in: "What are reasons not to have school year-round?" or "What are qualities of seals?" or "How do uniforms make school better?" A few asked for organizational support, as in: "Can you give me an outline for how I would write an argument about mascots?" Others asked for content support, as in: "Do you have any data points for this argument?" These prompts align with more than one aspect of planning. They are certainly requests for ideas. But the prompts also represent *goal-setting*; that is, the student aimed to build an argument with reasons and evidence.

Other prompts represented both *generating ideas* and *organizing*. Another student, also writing about school calendars, drafted a first paragraph on her Google Doc in which she argued against year-round school. After she completed this paragraph, she returned to ChatGPT to ask, "Can you also give me examples on why it would be beneficial to have school all year long?" This prompt for content also indicates that the student was drawing on some schema for the organization of an argument that included a counter argument.

Translating

Only sixteen percent of all prompts involved requests for ChatGPT to help students translate ideas to actual words or sentences on their Google Docs. Three pairs of students asked for compelling openers for an argument about mascots, as in: “Give examples of very persuasive opening statements about changing a school mascot.” However, as we will discuss, no students copied and pasted ChatGPT’s specific examples into their developing draft.

Reviewing

Thirty-one percent of prompts involved requests for reviewing, where students prompted ChatGPT in ways that helped them modify their drafts. For instance, after one student read through her draft, she asked ChatGPT to make her writing sound more fluid. One pair of students fed their complete argument draft to ChatGPT and prompted: “This is an argument on why we should change the school mascot to huskies, can you edit for grammar and coherency?” Another prompted, “Fix this essay.”

What students did not ask ChatGPT

It is worth noting that students did not use ChatGPT for one important aspect of planning: their initial claims and ideas. In building arguments about school mascots, for example, five out of six groups decided on a mascot before turning to ChatGPT. Only one pair considered asking ChatGPT about the best mascot for their school, but then decided that “GPT doesn’t know Sunrise,” so the bot would not be able to help them make that decision. Similarly, when writing arguments about year-round schooling or school uniforms, most students’ prompts indicated that they had decided on their position before beginning to build their argument. In one case, a student’s final piece indicated that they shifted their position after interacting with ChatGPT.

It is also worth noting that students used additional tools as they engaged in planning. One team used their school website to help complete a sentence about their school’s goals. Others interacted with Google Doc’s AI-driven autocomplete feature, sometimes accepting and sometimes ignoring the autocomplete suggestions. They also used Google’s spell-check feature. Finally, many students leaned on our written assignments for organizational support (“Let’s see...what else do we need to add?”). And perhaps most important, students used each other as supports in all aspects of the writing process. Even when students did not write with a partner, they asked questions of students sitting nearby, read over each

other’s shoulders, and laughed together at ChatGPT outputs.

How did students take up ChatGPT’s output?

In exploring how students adopted and rejected ChatGPT’s output, we used both quantitative and qualitative analysis.

Planning

Students used ChatGPT to plan their arguments by prompting the bot for data points or outlines. We wanted to know the extent to which students adopted ChatGPT’s output directly from those planning prompts. Using Jaccard similarity, we measured the percentage of individual words, two-word strings (e.g., “mascot represents”) and three-word strings common to ChatGPT’s output and students’ subsequent drafts (Table 2).

Given that ChatGPT output and student drafts addressed the same topics (e.g., school uniforms or mascots), a 17% similarity for individual words is not surprising. More important is that students’ drafts and ChatGPT outputs rarely contained the same two or three-word phrases. These results indicate that as students moved from planning to writing actual sentences, they rarely lifted language directly from ChatGPT. Instead, students were more likely to use ChatGPT’s output as a jumping off point for claims or propositions, as illustrated in Table 3.

Translating

Sixteen percent of students’ prompts asked ChatGPT to engage in *translation* (moving from ideas to specific words on the page), as when students prompted, “Give examples of very persuasive opening statements,” or “Give an example of a hook.” These requests seemed likely to lead to cutting and pasting. However, our transcripts and screencasts indicate that students did not copy from ChatGPT, but instead used ChatGPT’s examples as a model for their own translation. For instance, one pair of students—a junior and a sophomore—asked ChatGPT for examples of persuasive opening statements. ChatGPT offered five examples, including:

TABLE 2 Percentage of text shared by ChatGPT outputs and students’ essay drafts.

Individual words	Two-word strings	Three-word strings
17%	4%	1%

TABLE 3 Individual words and two- and three-word strings common to ChatGPT output and student drafts.

Turn	Speaker/writer	Content
1	Student prompts ChatGPT	<i>What are characteristics of a seal animal</i>
2	ChatGPT output	Characteristics of seals , the marine mammals, include: 1. Social behavior: Seals are social animals, sometimes living in large groups or colonies, and have complex communication systems that include sounds, gestures, and postures 2. Endurance : Seals have impressive endurance, allowing them to swim for hours or even days without rest 3. Playful behavior: Seals are known for their intelligent , curious , and playful behavior, often interacting with each other in acrobatic ways and playing with objects like rocks or seaweed
3	Students' developing draft	<i>Sunrise is a community where students strive for excellence in high school and further along in life. We believe that Seals should represent Sunrise's new mascot because of its values representing the school's community's endurance to stay attended in class, and their educational value. Seals are intelligent, curious, and resilient, foreshadowing the students at Sunrise because of their sacrifices for long school hours, curiosity about education, their intelligence and strive for success</i>

Note: Common words in bold.

1. "As we gather here today, I want to draw your attention to a matter that has been long overdue—the need to change our school mascot. Let's take this opportunity to show the world that we stand for inclusivity, diversity, and respect for all."
2. "Ladies and gentlemen, let's talk about the elephant in the room – our school mascot. It's time to retire the outdated, offensive imagery and come up with something that represents the values and beliefs of our student body today."

After reading the examples, the junior observed, "Okay, it's like a hook, like an address and a hook. And then a lot of [the sentences] open up with like...what's that rhetoric? The'y like use pathos to gather their audience. Like sympathy or empathy. You'll learn it next year."

Then the students toggled to their Google Doc. The junior asked the sophomore, "Okay, so what's our address to the audience going to be?" The students then wrote their own opening statement, which began: "My fellow Sunrisers, we understand the Panther has been Sunrise's pride and symbol since the beginning. But where do you see panthers here, except the zoo?"

Reviewing

When prompting ChatGPT in ways that would help students review their drafts, students sometimes prompted the bot to help them review a single choice of words (e.g., "What's a word with a more positive connotation which means the same as suitable?"). Other times—five instances total—students prompted ChatGPT to review their entire draft. They did so by copying the whole draft into ChatGPT. In response, ChatGPT functioned a bit like Grammarly or another editing application, returning their entire draft with changes in grammar, syntax, and language.

In these five instances, we manually compared students' drafts, ChatGPT's edits, and students' final arguments to see what students took up from the bot.

In the paired writing session, three out of six pairs of students asked ChatGPT to edit their entire draft. One pair of writers accepted all of ChatGPT's edits of their opening and closing paragraphs and rejected everything else. Another pair only accepted ChatGPT's edits of their opening paragraph.

The final pair asked ChatGPT to "proofread this argument." However, ChatGPT's edits were so minor—a few changes in punctuation—that the pair did not notice them. One student said to his partner, "It's the same." Then the students changed their prompt to ChatGPT, asking the bot to "fix this essay." This time, ChatGPT lagged; it offered no response. The students input their prompt two more times. Finally, one student said, "That's all the edits it wants to do." In this instance, we cannot know whether the students would have taken up more substantive edits.

In the individual writing session, two out of ten students asked ChatGPT to review and edit their arguments. One student accepted all of ChatGPT's changes, copying and pasting ChatGPT's edits into her Google Doc and submitting that version as her final draft. The other also accepted all of ChatGPT's edits and then added three more sentences of argument to ChatGPT's edits.

In all instances, students did not evaluate ChatGPT's edits sentence by sentence or accept one type of edit (e.g., punctuation) while rejecting other types of edits (e.g., diction). Adopting ChatGPT's edits was more of an all-or-nothing affair.

Two cases of students using ChatGPT as a writing resource

The previous findings represent analysis based on tallies of individual discursive moves or units

of language (e.g., words or paragraphs). While informative, those measures cannot account for the complex and iterative nature of the writing process. To generate more nuanced insights into students' use of ChatGPT, we present two cases of students in initial and later stages of composing their arguments about school mascots.

Case 1: Initial planning, translating, and reviewing

As we noted earlier, we coded most of students' written requests of ChatGPT as belonging to the *planning* process (e.g., "Can you tell me the qualities of a warrior?"). However, students wove ChatGPT-supported planning into all other writing processes. A closer look at partners Homero and Alex (pseudonyms) highlights those processes.

Homero and Alex began their mascot argument by pitching possible mascots to one another. Alex first suggested a mongoose ("They eat snakes!") or a titan. Ultimately, they settled on a warrior as a good symbol of Sunrise High School. Then, having made this choice,

they asked the bot to help them generate further ideas (Table 4).

Although students' input to ChatGPT mostly focused on the process of planning their arguments, students used ChatGPT's output to cycle through all writing processes in quick succession. For instance, after deciding on a mascot, Homero and Alex used their input to ChatGPT to engage in some organizing and goal-setting; that is, they knew that to further their argument, they would first need to describe characteristics of a warrior (turns 3 and 4). Then, when ChatGPT responded to their prompt (turn 5), Homero and Alex moved to *reviewing*; specifically, they decided which aspects of ChatGPT's output would address the rhetorical challenge of matching a mascot to their school. They identified qualities like resilience and discipline as relevant to their argument, but they rejected qualities like courage and loyalty (turns 7–12). Then they moved to *translating* their selection of characteristics into a proposition, writing "[Warriors] are disciplined, which ties to Sunrise in how..."

TABLE 4 Students Homero and Alex interact with each other and ChatGPT.

Turn of talk	Speaker/writer	Students' talk (Students' writing in italics)
1	Homero	All right, low key, just Warriors. Let's get started, please.
2	Homero types into google doc	<i>We believe the current mascot of Sunrise doesn't reflect Sunrise's goal as a school and as a community.</i>
3	Alex	So we need the characteristics of...
4	Homero asks ChatGPT	<i>What are characteristics of a warrior</i>
5	GPT generates output	Some of the common characteristics of a warrior are: 1. Courage: A warrior is brave in the face of danger and is willing to take risks to protect their beliefs and values. 2. Discipline: A warrior has a strong sense of self-discipline, adheres to strict codes of conduct, and is highly skilled. 3. Resilience: A warrior bounces back from setbacks and perseveres in the face of adversity. 4. Loyalty: A warrior is fiercely loyal to their family, community, and values. 5. Strength: A warrior is physically and mentally strong, with endurance to overcome challenges. 6. Honor...
6	Alex reads from ChatGPT output	"Courage...discipline...resilience..."
7	Homero	I think resilience and then discipline tie in with Sunrise.
8	Alex	Honor—what about honor?
9	Homero	Honor too.
10	Alex	Strength? Like mentally strong?
11	Homero	Yeah, strength.
12	Alex	Discipline, resilience, honor. Of course intelligence, but we don't have to put that.
13	Homero types into google doc	<i>A mascot that would best reflect Sunrise's mission would be the Warriors. They are disciplined, which ties to Sunrise in how [stops writing].</i>
14	Homero toggles from google doc back to ChatGPT output and reads	"Discipline...highly trained in their skills." Now how does that tie into Sunrise? I think the highly trained...
15	Alex	I was going to name each characteristic and just say that.
16	Homero types into google doc	<i>They are disciplined, resilient, honorable, and adaptable. Sunrise students are need these traits on a daily basis to thrive.</i>

The two students then moved back to *planning* their argument (“I was just going to name each characteristic” in turn 15) and then to *translating* their selection of ChatGPT’s output into a detailed claim about warriors (“They are disciplined, resilient, honorable and adaptable” in turn 16).

This excerpt of Homero and Alex’s dialogue corroborates the results of our quantitative analysis: When planning their arguments, students did not copy and paste ChatGPT’s language into their arguments. They evaluated the salience of ChatGPT’s output, using their own language to translate ideas into sentences, including changing ChatGPT’s nouns to adjectives to compose an efficient claim. In other words, they did not use the chatbot as a cheatbot.

Case 2: Subsequent planning, translating, and reviewing

When students moved from initial planning and translating to reviewing their complete drafts, they tended to adopt more of ChatGPT’s suggestions. For instance, Daisy and Chloe (pseudonyms) developed an argument in favor of a husky as the best mascot for their school. After about 20 min of prompting ChatGPT, talking with one another, and learning the word “felicitous” from ChatGPT, Daisy and Chloe had drafted a complete argument.

The pair then decided to paste their entire argument into ChatGPT. They gave context for their argument to

ChatGPT and asked the bot to “edit for grammar and coherency.” Below are the students’ complete draft (Table 5) and ChatGPT’s edited version of the students’ draft (Table 6).

As Daisy and Chloe read ChatGPT’s edited version of their argument, they began the dialogue below (Table 6).

At this stage, Daisy and Chloe took up ChatGPT’s output to engage in several subprocesses of *reviewing*: First, they *evaluated* ChatGPT’s output, both on its own and in relation to the strengths of their original writing. They found ChatGPT’s introductory paragraph to be “worded better” than their version (turn 4). At the same time, they noted that ChatGPT’s version of their argument “took out all of our personality!” (turn 4). Their evaluations led to *revising* decisions; for instance, Daisy said, “Okay, this one, the ending sentence...we could use their ending” (turn 10). Ultimately, Daisy and Chloe adopted all of ChatGPT’s first and last paragraph, but kept their version of the body of their argument, including the playful references to hippies. The students made no other changes to the body of their argument.

Daisy and Chloe may have also taken up ChatGPT’s output in less obvious ways. For instance, note that the pair wrote their initial argument as one long paragraph. When ChatGPT edited their draft, the bot broke the argument into four paragraphs. Though the students did not comment on ChatGPT’s paragraph breaks, they

TABLE 5 Daisy and Chloe ask ChatGPT to edit their draft.

Daisy and Choe's draft	GPT's edit
Just one of the reasons that Sunrise is special is that it is a dog friendly campus, we feel this is now a part of the Sunrise culture and it should be reflected in the mascot. Most notably the dog who has been here the longest, Rocky the husky (Helen's dog). At Sunrise we pride ourselves in being intelligent, strong, loyal, and independent. We believe that a Husky would be a felicitous fit because they also represent these same qualities. Although the other options can be suitable for other schools, they really don't align with ours. First off, a redwood tree?? Terrible mascot, not hate to Stanford (they make it work), but in a high school environment this could weaken school pride. Come on what highschoolers would want to say GO TREES!! (a hippie??) Additionally, they don't truly represent as much of our desired qualities as Sunrise students. The other option of a Seal as a mascot, may originally seem fun and creative, but we are strong and fierce students, and well that's not really what comes to mind when thinking of a seal. The final option was Warriors, and while this one might sound strong, if you think about it, it gives off the idea that we are violent, which is not something that is acceptable at Sunrise. We are also very inclusive at Sunrise, and a Warrior could be seen as insensitive towards some groups. Overall, we should pride ourselves in having a mascot which mirrors our strengths as students, along with having that special connection of dogs on campus. If you ask anyone on campus, they will say something positive about Rocky and what better mascot, then one that everyone loves already?	One of the things that sets Sunrise apart is our dog-friendly campus, and we believe that this should be reflected in the school's mascot. For many of us, the longest-standing dog on campus is Helen's husky, Rocky. Sunrise prides itself on being intelligent, strong, loyal, and independent, and we believe that huskies embody these same qualities While other mascots may be suitable for other schools, they do not align with the values and qualities we have as Sunrise students. The idea of a redwood tree as a mascot seems terrible and would be inappropriate for a high school setting. It could also weaken the school's spirit and pride. Similarly, seals may be fun and creative, but they do not represent the strength and fierceness that we have as students The third option of Warriors may sound strong, but it could be seen as glorifying violence which is against the inclusive and accepting nature of Sunrise. We also want to reinforce the importance of being sensitive towards all cultural groups Overall, we need a mascot that reflects the strengths and values of Sunrise. Huskies represent exactly that, and they already have a special connection with our campus. The appeal of Rocky is just one of the things that showcase the excellent qualities of huskies. If you ask any student about Rocky, you'll find they have something positive to say. With Sunrise being a dog-friendly campus, choosing huskies as our mascots seems like a felicitous fit

TABLE 6 Chloe and Daisy's dialogue during the reviewing process.

Turn of talk	Speaker	Students' talk
1	Daisy reads ChatGPT's output on the computer screen	Dang, we wrote a lot!
2	Chloe points cursor at first paragraph	Oh my gosh, this is way better.
2	Daisy	Way better. Dang. And this part [the introduction] is kind of fun.
3	Chloe	Okay, I like the introduction for this. So I think we could switch [our introduction for ChatGPT introduction] because it's saying the same thing, but it's more like...
4	Daisy	Worded better. It took out all of our personality, though!
5	Chloe	Did it take out the hippie?
6	Daisy	It took out the hippie!
7	Chloe	I don't know... I like our arguments better than theirs.
8	Daisy	Okay, this one, the ending sentence...we could use their ending.
9	Chloe	Yeah, the conclusion.

did incorporate paragraphs into their final submission. In this way, they drew on ChatGPT not just for revisions to grammar and diction, but for structure.

The students also seemed to take up a meta-level awareness of their writing style and voice after reading ChatGPT's output. Chloe and Daisy recognized the difference between ChatGPT's blandly written argument and their own writing, with its playful tone and "personality." Without ChatGPT's contrasting case, the students may not have recognized the value of that personality.

Limitations

To make the study manageable for students, we limited each writing session to one hour, which meant that during the partnered writing session, students had about 50 minutes to compose their arguments; during the individual sessions, students had about 30 minutes. It seems likely that, if each session had been longer, at least a few more students might have asked ChatGPT to edit their work. It is also possible that, given more time, students might have been more selective about choosing which edits to accept. Ideally, future studies will allow more time for writing.

DISCUSSION

Writing teachers and scholars understand that writing is an unsequestered, social process. Planning, translating, and evaluating do not take place only in a student's head. Those processes are distributed across teachers, peers, settings, and technology, among other things. Our study indicates that ChatGPT can become a part of that unsequestered model of writing, and that students can use ChatGPT as a resource for writing without sidestepping the processes of planning, translating, and evaluating. In fact, we found that for

the most part, participating students treated ChatGPT as a stand-in for a teacher, peer, or range of existing online sources and digital applications. They asked ChatGPT for planning assistance, including ideas and definitions, just as they might do with a teacher or Google search.

Similarly, they used ChatGPT for translation and reviewing assistance just as they might use Grammarly or other editing tools. In Puentedura's (2013) terms, they used ChatGPT as a substitute for other, existing writing supports, as opposed to using ChatGPT to modify or redefine the writing process. If more students had been more familiar with the tool, it is possible that they might have used it more expansively. As students do become more familiar with text-based AI, and as AI becomes even more sophisticated, students may indeed redefine their writing practices. We will then have to revisit our concerns about ways that chatbots may cheat students out of learning to write. But in our study, we did not see that concern play out. In fact, ChatGPT may have enhanced students' engagement in planning, translating, and reviewing in several ways. Below, we outline some of those ways, and offer suggestions for educators as they move to integrate ChatGPT into their understanding of the writing process.

Planning

In much writing instruction, both pre- and post-ChatGPT, some of students' goal-setting and organizing is distributed to teachers' assignments or rubrics. For instance, our mascot argument assignment called for claims and evidence, so students did not have to develop those writing goals themselves. Thus it was not surprising that students' prompts to ChatGPT included requests for claims and evidence, such as "What would make warriors a good mascot?" or "Can you give me some data?"

However, students' prompts to ChatGPT did not only follow the goals of our assignment. Their prompts also included requests such as, "What are the symbolic qualities of huskies?" or "What are common held preconceptions about warriors?" As we see it, articulating these prompts required students to consider argument form, structure, and audience; in other words, students had to plan in order to prompt. In this way, instead of helping students disengage from planning processes, ChatGPT may offer students the opportunity to practice goal-setting. Teachers will play a key role in helping students become aware of the planning behind their prompts, and thus deepen their meta-level knowledge of writing. Building this awareness will also require teachers to become more familiar with AI prompting to help students craft and analyze the results of their prompts.

Translating

In translating ideas to words on the page, text-based AI may offer students an opportunity that other writing applications do not yet offer, and that teachers cannot reasonably offer: multiple, tailored, contrasting translations of the same idea. From these contrasting cases, students can develop meta-level understandings of rhetorical choices and effects. We saw this meta-level understanding emerge as one pair of students examined several ChatGPT-generated opening statements for the mascot argument. The students were able to identify the common features of the opening statements: "A lot of [the sentences] open up with like...what's that rhetoric? They, like, use pathos

to gather their audience. Like sympathy or empathy." Then the students wrote an opening statement modeled on those features.

Such tailored contrasting cases can be a robust learning tool (Mah & Levine, 2023; Schwartz & Bransford, 1998), but teachers often do not have time to generate such cases, and other digital tools are not yet able to do so. This feature of text-based AI thus opens new doors for teaching.

Reviewing

Arguably, students treated ChatGPT most like a cheatbot when reviewing their work. After asking ChatGPT to "fix" or "correct" their work, students often took up ChatGPT's suggestions wholesale, sometimes accepting ChatGPT's edits of whole paragraphs. To address this issue, teachers will need to guide students to carefully compare their drafts to ChatGPT's edits to identify and evaluate the underlying rules that ChatGPT seems to follow. We see this kind of comparative work as a useful way to teach about grammar, syntax, punctuation, and revision. Doing so may give students much-needed practice with revision (Kihara et al., 2009).

When students did compare their writing with ChatGPT's, they were not only able to identify ChatGPT's moves, but recognize the distinctiveness of their own voice and rhetorical moves. For instance, when Chloe and Daisy contrasted their original language with ChatGPT's revision, they noticed that ChatGPT "took out all of our personality!" ChatGPT's bland revision may have acted as a contrasting case for the students, allowing them to identify their own

TABLE 7 Making writing processes visible through ChatGPT.

Process	Learning goals	Example from our data	Teacher questions to help make writing processes visible
Planning	Help students identify the structure of a genre or mode of writing (such as argument) based on their prompts to the chatbot	When writing an argument about a new school mascot, one pair of students asked ChatGPT, "Why would trees be a weak school mascot?"	"Can you explain why you asked that question of the chatbot?" "Can you talk about how including this data might strengthen an argument like this?"
Translating	Help students learn to ask chatbot for several examples (not just one), and then compare examples to identify underlying principles	Students asked ChatGPT to "give four examples of very persuasive opening statements."	"What do those persuasive statements have in common?" "What are some of the principles the chatbot seems to be following in making these persuasive statements?" "Let's try writing a similar kind of opening for another argument."
Reviewing	Help students understand why they think the chatbot's edits make their writing stronger, weaker, or both	Students decided that they liked ChatGPT's edited version of their introduction "because it's saying the same thing...but it's more like...worded better."	"What exactly makes it 'worded better,' in your mind? And worded better for whom, do you think?" "Where do you think the chatbot didn't improve your writing, and why?"

livelier voice and tone. Without the bot's contrasting example, they might not have noted their skill.

Making thinking visible will help students learn from text-based AI

Our findings suggest that teachers can facilitate responsible, productive use of ChatGPT and other text-based AI across all writing processes. One powerful way to facilitate is to help make visible the instincts, knowledge, and goals that underlie students' input to text-based AI and their adoption of AI's outputs. Doing so may not only help students learn more about writing processes, but help them embrace writing as a social, distributed activity. In Table 7, we offer examples of questions teachers might ask to make students' thinking visible at different stages of the writing process.

CONCLUSION

We do not wish to minimize teachers' legitimate fears around cheating. However, our study offers a glimpse of how students can use AI tools productively and responsibly, in ways that align with widely-adopted educational frameworks such as that of the International Society for Technology in Education, which calls for students to "use technology to seek feedback that informs and improves their practice" and "critically curate a variety of resources using digital tools to construct knowledge" (ISTE, 2024). Teachers may find it more productive and fulfilling to focus their energy on guiding students to use generative AI responsibly, as opposed to attempting the difficult task of detecting cheating—or even defining it—in this new era of AI (Dehouche, 2021). We hope our findings help teachers and researchers explore the affordances of AI to make writing a more social, unsequestered activity.

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CONFLICT OF INTEREST STATEMENT

None.

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